

ENTERPRISE NETWORK ENGINEERING

Enterprise Campus Network: Multi-Branch Infrastructure & Engineering Implementation

IMPLEMENTATION & ENGINEERING DOCUMENTATION • CISCO INFRASTRUCTURE

LAYER 2 • Rapid-PVST+ / LACP / VLANs	LAYER 3 • OSPF Area 0 / Inter-VLAN / DHCP Relay	UNIFIED COMMS • Cisco CME / 7960 Phones / Option 150
--------------------------------------	---	--

ENGINEERING TEAM Bishoy Gerges & Malak Abdelaziz	DOCUMENT VERSION & STATUS 1.0.2026 • Final Verified CLI Build	SCOPE Two-Branch Enterprise Campus Network
---	--	---

1. EXECUTIVE SUMMARY & PROJECT OBJECTIVES

This project delivers an enterprise-grade campus network architecture spanning two primary functional branches (Upper and Lower office blocks) interconnected via Cisco 2811 Integrated Services Routers (ISRs). The infrastructure integrates high-availability Layer 2 switching, dynamic routing, centralized network management services, Quality of Service (QoS) enforcement, and unified VoIP telephony.

ENGINEERING GOAL	IMPLEMENTATION DETAILS
Loop-Free Resilient Switching	Rapid-PVST+ with deterministic bridge priority tuning (28672) and IEEE 802.3ad LACP link aggregation.
Traffic Isolation & Segmentation	Separation of corporate departmental data VLANs 10–60 from VoIP VLANs 110–160.
Unified Communications (VoIP)	On-premise IP PBX telephony using Cisco CallManager Express (CME) with automated Option 150 TFTP endpoint provisioning.
QoS Bandwidth Optimization	CBWFQ allocating 30% strict-priority bandwidth for voice streaming.
Centralized Network Services	High-availability DHCP relay architecture consolidating network parameters onto management server 192.168.1.2.
Boundary Security	Granular extended ACLs isolating sensitive server segments and blocking unauthorized remote administrative access.

2. LOGICAL TOPOLOGY OVERVIEW

UPPER OFFICE BLOCK - Access / Distribution Switching - Data VLANs 10–60 - Voice VLANs 110–160	CISCO 2811 ISRS (CORE / INTERCONNECT) - Branch / Inter-VLAN Router: R1 (Branch-Router) - Central CME PBX: R2 (CME-Voice-Router) - OSPF Area 0 Dynamic Routing • Central Mgmt: 192.168.1.2	LOWER OFFICE BLOCK - Access / Distribution Switching - Data VLANs 10–60 - Voice VLANs 110–160
---	--	---

3. IP ADDRESSING & VLAN ALLOCATION MATRIX

Data DHCP Pools & Subnet Allocation (Server 192.168.1.2)

VLAN	SUBNET NAME / PURPOSE	NETWORK ADDRESS	SUBNET MASK	USABLE HOST RANGE	GATEWAY
10	Department 1 - Data	192.168.10.0	255.255.255.128 (/25)	192.168.10.2 – 192.168.10.126	192.168.10.1
20	Department 2 - Data	192.168.20.0	255.255.255.192 (/26)	192.168.20.2 – 192.168.20.62	192.168.20.1
30	Department 3 - Data	192.168.30.0	255.255.255.0 (/24)	192.168.30.2 – 192.168.30.254	192.168.30.1
40	Department 4 - Data	192.168.40.0	255.255.255.240 (/28)	192.168.40.2 – 192.168.40.14	192.168.40.1
50	Department 5 - Data	192.168.50.0	255.255.255.128 (/25)	192.168.50.2 – 192.168.50.126	192.168.50.1
60	Department 6 - Data	192.168.60.0	255.255.255.192 (/26)	192.168.60.2 – 192.168.60.62	192.168.60.1

Voice & Management Subnets

VLAN	SUBNET NAME / PURPOSE	NETWORK ADDRESS	SUBNET MASK	TFTP SERVER (OPT 150)	GATEWAY
110	Voice Dept 1	192.168.110.0	255.255.255.240 (/28)	192.168.1.1	192.168.110.1
120	Voice Dept 2	192.168.120.0	255.255.255.240 (/28)	192.168.1.1	192.168.120.1
130	Voice Dept 3	192.168.130.0	255.255.255.240 (/28)	192.168.1.1	192.168.130.1
140	Voice Dept 4	192.168.140.0	255.255.255.240 (/28)	192.168.1.1	192.168.140.1
150	Voice Dept 5	192.168.150.0	255.255.255.240 (/28)	192.168.1.1	192.168.150.1
160	Voice Dept 6	192.168.160.0	255.255.255.240 (/28)	192.168.1.1	192.168.160.1
Core	Core Transit & Servers	192.168.1.0	255.255.255.0 (/24)	N/A	192.168.1.1

4. INTER-BRANCH & SERVICES ROUTER (R1 - BRANCH / INTER-VLAN ROUTER)

Comprehensive Cisco IOS running configuration establishing sub-interface IEEE 802.1Q encapsulation, QoS traffic policing, dynamic OSPF Area 0 routing, and extended boundary security ACLs:

DEVICE: Branch-Router (Cisco 2811 ISR)

Cisco IOS CLI

```
!  
hostname Branch-Router  
!  
ip cef  
no ipv6 cef  
!  
spanning-tree mode pvst  
!  
class-map match-any voice  
  match access-group name voice-traffic  
!  
policy-map voip-qos  
  class voice  
    priority percent 30  
  class class-default  
    fair-queue  
!  
interface FastEthernet0/0  
  no ip address  
  service-policy output voip-qos  
  duplex auto  
  speed auto  
!  
interface FastEthernet0/0.10  
  encapsulation dot1Q 10  
  ip address 192.168.10.1 255.255.255.128  
  ip helper-address 192.168.1.2  
!  
interface FastEthernet0/0.20  
  encapsulation dot1Q 20  
  ip address 192.168.20.1 255.255.255.192  
  ip helper-address 192.168.1.2  
  ip access-group block-vlan20 in  
!  
interface FastEthernet0/0.30  
  encapsulation dot1Q 30  
  ip address 192.168.30.1 255.255.255.0  
  ip helper-address 192.168.1.2  
  ip access-group block-vlan30-ssh in  
!  
interface FastEthernet0/0.40  
  encapsulation dot1Q 40  
  ip address 192.168.40.1 255.255.255.240  
  ip helper-address 192.168.1.2  
!  
interface FastEthernet0/0.50  
  encapsulation dot1Q 50  
  ip address 192.168.50.1 255.255.255.128  
  ip helper-address 192.168.1.2  
!  
interface FastEthernet0/0.60  
  encapsulation dot1Q 60  
  ip address 192.168.60.1 255.255.255.192  
  ip helper-address 192.168.1.2  
!  
interface FastEthernet0/0.110  
  encapsulation dot1Q 110  
  ip address 192.168.110.1 255.255.255.240  
  ip helper-address 192.168.1.2  
!  
interface FastEthernet0/0.120  
  encapsulation dot1Q 120  
  ip address 192.168.120.1 255.255.255.240  
  ip helper-address 192.168.1.2  
!  
interface FastEthernet0/0.130  
  encapsulation dot1Q 130  
  ip address 192.168.130.1 255.255.255.240  
  ip helper-address 192.168.1.2  
!  
interface FastEthernet0/0.140  
  encapsulation dot1Q 140  
  ip address 192.168.140.1 255.255.255.240  
  ip helper-address 192.168.1.2  
!  
interface FastEthernet0/0.150  
  encapsulation dot1Q 150  
  ip address 192.168.150.1 255.255.255.240  
  ip helper-address 192.168.1.2  
!  
interface FastEthernet0/0.160  
  encapsulation dot1Q 160
```

```

ip address 192.168.160.1 255.255.255.240
ip helper-address 192.168.1.2
!
interface FastEthernet0/1
ip address 192.168.1.3 255.255.255.0
ip helper-address 192.168.1.2
service-policy output voip-qos
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 192.168.1.0 0.0.0.255 area 0
network 192.168.10.0 0.0.0.127 area 0
network 192.168.20.0 0.0.0.63 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.15 area 0
network 192.168.50.0 0.0.0.127 area 0
network 192.168.60.0 0.0.0.63 area 0
network 192.168.110.0 0.0.0.15 area 0
network 192.168.120.0 0.0.0.15 area 0
network 192.168.130.0 0.0.0.15 area 0
network 192.168.140.0 0.0.0.15 area 0
network 192.168.150.0 0.0.0.15 area 0
network 192.168.160.0 0.0.0.15 area 0
!
ip classless
!
ip access-list extended block-vlan20
deny ip 192.168.20.0 0.0.0.63 192.168.1.0 0.0.0.255
permit ip any any
!
ip access-list extended block-vlan30-ssh
deny tcp 192.168.30.0 0.0.0.255 192.168.1.0 0.0.0.255 eq 22
permit ip any any
!
ip access-list extended voice-traffic
permit ip 192.168.110.0 0.0.0.15 any
permit ip 192.168.120.0 0.0.0.15 any
permit ip 192.168.130.0 0.0.0.15 any
permit ip 192.168.140.0 0.0.0.15 any
permit ip 192.168.150.0 0.0.0.15 any
permit ip 192.168.160.0 0.0.0.15 any
permit ip 192.168.1.0 0.0.0.255 192.168.110.0 0.0.0.15
permit ip 192.168.1.0 0.0.0.255 192.168.120.0 0.0.0.15
permit ip 192.168.1.0 0.0.0.255 192.168.130.0 0.0.0.15
permit ip 192.168.1.0 0.0.0.255 192.168.140.0 0.0.0.15
permit ip 192.168.1.0 0.0.0.255 192.168.150.0 0.0.0.15
permit ip 192.168.1.0 0.0.0.255 192.168.160.0 0.0.0.15
!
line con 0
line aux 0
line vty 0 4
login
!
end

```

5. CENTRAL UNIFIED COMMUNICATIONS PBX (R2 - CISCO CME ROUTER)

Central PBX setup provisioning 12 distinct Cisco 7960 IP Phones across the dual-branch topology with complete MAC binding and directory number mapping, utilizing Central Voice Router (192.168.1.1) as the CallManager Express (CME) PBX:

DEVICE: CME-Voice-Router (Cisco CallManager Express)

Cisco IOS CLI

```
!  
hostname CME-Voice-Router  
!  
ip cef  
no ipv6 cef  
!  
spanning-tree mode pvst  
!  
interface FastEthernet0/0  
ip address 192.168.1.1 255.255.255.0  
duplex auto  
speed auto  
!  
interface FastEthernet0/1  
no ip address  
shutdown  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 192.168.1.0 0.0.0.255 area 0  
!  
ip classless  
!  
telephony-service  
max-ephones 20  
max-dn 20  
ip source-address 192.168.1.1 port 2000  
auto assign 1 to 10  
auto assign 7 to 12  
!  
ephone-dn 1  
number 1001  
!  
ephone-dn 2  
number 1002  
!  
ephone-dn 3  
number 1003  
!  
ephone-dn 4  
number 1004  
!  
ephone-dn 5  
number 1005  
!  
ephone-dn 6  
number 1006  
!  
ephone-dn 7  
number 2001  
!  
ephone-dn 8  
number 2002  
!  
ephone-dn 9  
number 2003  
!  
ephone-dn 10  
number 2004  
!  
ephone-dn 11  
number 2005  
!  
ephone-dn 12  
number 2006  
!  
ephone 1  
device-security-mode none  
mac-address 0090.21EA.48B7  
type 7960  
button 1:1  
!  
ephone 2  
device-security-mode none  
mac-address 0030.F206.77E3  
type 7960  
button 1:2  
!
```

```
ephone 3
 device-security-mode none
 mac-address 0040.0BBB.D192
 type 7960
 button 1:3
!
ephone 4
 device-security-mode none
 mac-address 0007.EC52.DA5C
 type 7960
 button 1:4
!
ephone 5
 device-security-mode none
 mac-address 0040.0B85.92E5
 type 7960
 button 1:5
!
ephone 6
 device-security-mode none
 mac-address 00D0.BC90.3CD7
 type 7960
 button 1:6
!
ephone 7
 device-security-mode none
 mac-address 0004.9A6C.25EE
 type 7960
 button 1:7
!
ephone 8
 device-security-mode none
 mac-address 0001.C93C.E2AC
 type 7960
 button 1:8
!
ephone 9
 device-security-mode none
 mac-address 0007.EC69.8E58
 type 7960
 button 1:9
!
ephone 10
 device-security-mode none
 mac-address 0060.4741.513A
 type 7960
 button 1:10
!
ephone 11
 device-security-mode none
 mac-address 0060.2F43.ECC9
 type 7960
 button 1:11
!
ephone 12
 device-security-mode none
 mac-address 0003.E489.44AE
 type 7960
 button 1:12
!
line con 0
line aux 0
line vty 0 4
 login
!
end
```

6. LAYER 2 INFRASTRUCTURE SWITCHING CONFIGURATIONS

Core / Distribution Layer Switching (Root Bridge & EtherChannel Trunking)

Deterministic STP root configuration across enterprise VLANs with 802.3ad LACP port-channel trunks:

DEVICE: Core-Dist-Switch (Cisco Catalyst 3560/2960)

Cisco IOS CLI

```
!
hostname Core-Dist-Switch
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
spanning-tree vlan 10,20,30,40,50,60,110,120,130,140,150,160 priority 28672
!
interface Port-channel2
 switchport mode trunk
!
interface range FastEthernet0/1-7, FastEthernet0/9-10
 switchport mode trunk
!
interface range FastEthernet0/14-15
 switchport mode trunk
 channel-group 2 mode active
!
interface Vlan1
 no ip address
 shutdown
!
line con 0
line vty 0 4
 login
line vty 5 15
 login
!
end
```

Edge / Access Layer Switching (Workstations, VoIP Drops & PortFast)

Access port assignment with multi-VLAN auxiliary voice tagging and Spanning-Tree PortFast for instantaneous edge forwarding:

DEVICE: Access-Switch-Drop (Cisco Catalyst 2960)

Cisco IOS CLI

```
!
hostname Access-Switch-Drop
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface range FastEthernet0/1-2
 switchport mode access
 switchport access vlan 10
!
interface FastEthernet0/3
 switchport mode trunk
!
interface FastEthernet0/4
 switchport mode access
 switchport access vlan 10
 switchport voice vlan 110
 spanning-tree portfast
!
interface Vlan1
 no ip address
 shutdown
!
line con 0
line vty 0 4
 login
line vty 5 15
 login
!
end
```

7. VERIFICATION & ACCEPTANCE TESTING

VERIFICATION COMMAND	EXPECTED OUTPUT	STATUS
show spanning-tree brief	Rapid-PVST+ root priority verified (28672)	PASSED / DETERMINISTIC
show etherchannel summary	Po2(SU) with members Fa0/14(P) & Fa0/15(P)	OPERATIONAL (LACP)
show ip route ospf	Full routing table populated via Area 0	CONVERGED
show policy-map interface Fa0/0	Priority queue packet matches on VoIP ACL	QOS ACTIVE (30% BW)
Ping End-to-End	0% packet loss (ICMP Reply Received)	PASSED
Ping Security Verification	VLAN 20 → 192.168.1.0/24 Destination Unreachable	ACCESS-LIST BLOCKED

✓ OVERALL ACCEPTANCE STATUS: PASSED

Core switching, dynamic routing, QoS priority queueing, telephony binding, and ACL verification tests completed successfully.